



perioperative nursing care





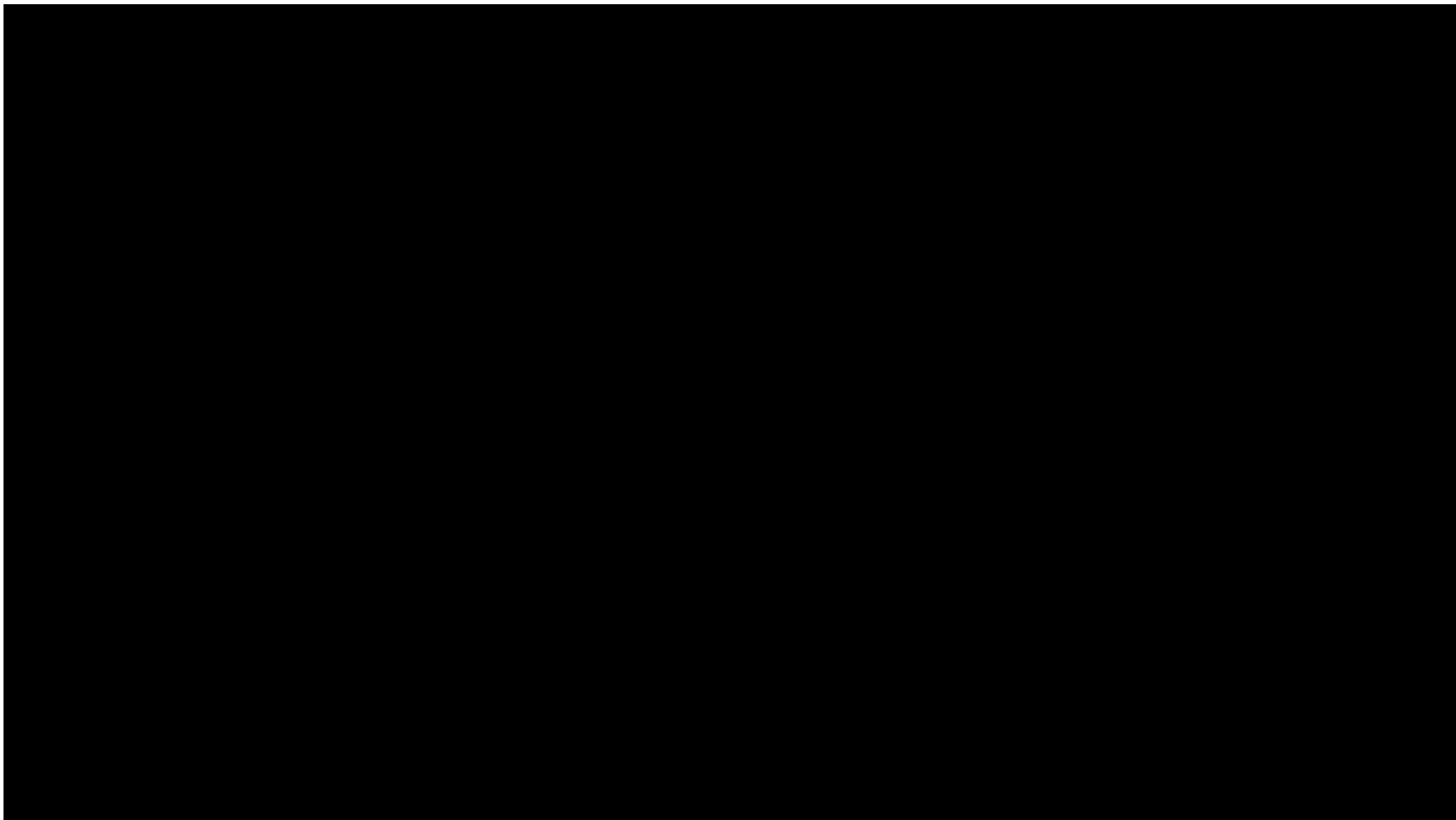
Learning objectives

1. Explain the **preparation of patients physically, emotionally, and psychosocially** for surgical procedures.
2. Identify the **types of patients most at risk** for surgical **complications**, and state why each patient is at risk.
3. **Plan and implement patient and family teaching** to prevent postoperative complications.
4. Compare the roles of the scrub nurse and the **circulating nurse**.
5. Analyze the differences in the various **types of anesthesia**
6. Perform a thorough **nursing assessment for a preoperative patient**.
7. Teach patient **postoperative exercises** during the preoperative period.
8. Prepare a **patient for surgery using a preoperative checklist**.

Teaching and Learning Method

1. Lecture
2. Discussion
3. Written assignments
4. Case study





Outlines:

Introduction

Preoperative care

Intraoperative care

Post-operative care



An introduction

Surgery is the use of instruments during an operation to treat injuries, diseases, and deformities.

Perioperative nursing refers to care of the patient before, during, and after surgery.

Surgery is done for several reasons



Preventive surgery removes tissue before it causes a problem as in mole to prevent cancer development.



Diagnostic, or exploratory, surgery takes tissue samples for study to make a diagnosis



Curative surgery involves the removal of diseased or abnormal tissue as in an inflamed appendix



Palliative surgery is done when an underlying condition cannot be corrected but symptoms need to be alleviated.



Cosmetics, or reconstructive: surgery is done to improve appearance

**Surgery according
to Urgency Level**

Emergent: Immediate surgery needed to save life

Urgent: Surgery needed within 24–30 hours

Elective: Planned/scheduled, with no time requirements

Surgery according to Degree of Risk of Surgery

Minor Procedure
without
significant
risk; often
done with
local anesthesia

Major Procedure: greater
risk, usually longer and
more extensive than a
minor procedure

Bloodless surgery.

Ambulatory surgery

includes outpatient, same-day, or short-stay surgery that does not require an overnight hospital stay but may entail an admission to an inpatient hospital setting for less than 24 hours

**A client admitted
for ambulatory
surgery must meet
the following
criteria:**

The client is not critically ill.

The surgical procedure does not require many hours of general anesthesia.

Recovery is expected to be quick, with minimal specialized care after surgery.

The client or family can provide adequate postoperative care.

Emergency Surgery

- ☞ Emergency surgeries **are unplanned and occur with little time for preparation** for the patient or the perioperative team
 - ☞ Preparation of the patient is essential , condensed time frame
 - ☞ **Informed consent and essential information**, such as pertinent past medical history and allergies, need to be obtained from a family member
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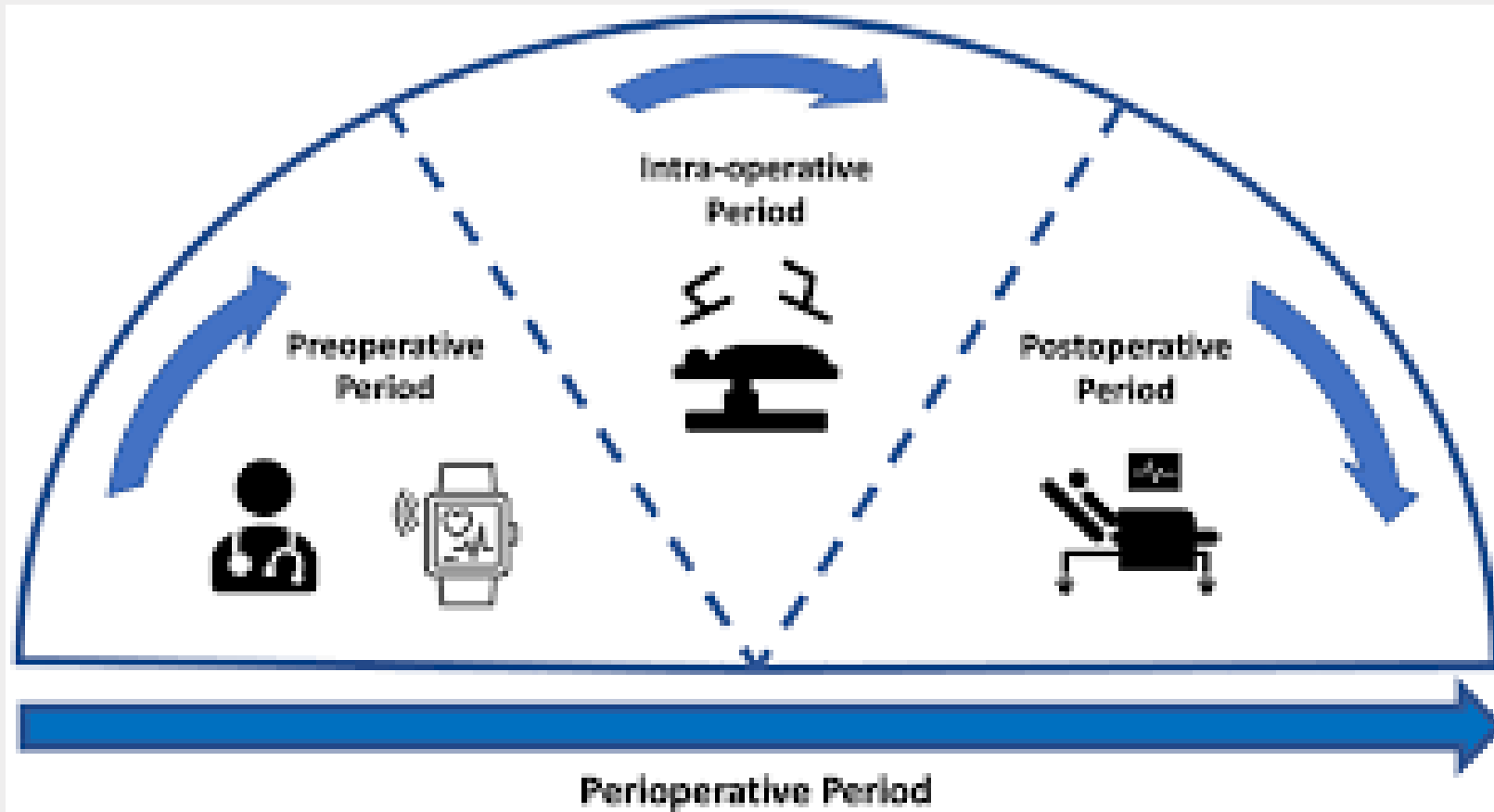
Perioperative Surgical Phases:

Perioperative: All three phases surrounding and during surgery

Preoperative: Begins with decision for surgery and ends with transfers to the operating room

Intraoperative: Begins with transfer to operating room and ends with admission to post-anesthesia care unit (PACU)

Postoperative: Begins with admission to PACU and continues until recovery is complete



1.Preoperative Phase

The primary
roles in
preoperative
phase are to:

- Assist in data collection for developing the patient's plan of care.
- Reinforce explanations and instructions given to the patient and family
- Provide emotional and psychological support

Factors Influencing Surgical Outcomes

Hydration and Nutrition

- A normal fluid and electrolyte balance decreases complications. Patients
- should be well nourished to adequately heal and recover from surgery

Smoking and Alcohol

- Smoking thickens and increases the amount of lung secretions

Diseases

- Chronic disorders increase the patient 's surgical risk unless they are well controlled.

Preoperative teaching:

All surgical patients should receive information related to:

- **Preoperative procedures:** Enemas, skin preparation, restriction of food and liquid intake, and administration of bedtime sedatives and preoperative medication; time to come to the hospital
- **Technical information:** Anticipated surgical procedure; location of incisions; dressings, tubes, drains, catheters, or other equipment that is expected
- **Day of surgery:** Time surgery is scheduled; time to arrive at the hospital or leave room, probable length of procedure, effects of preoperative medications, where family will wait, when and where family can see the patient after surgery, pain control, and postoperative routine

- **Post-anesthesia care unit (PACU):** General environment (noise, lights, equipment), frequent taking of vital signs, pulse oximetry, and administration of oxygen
- **Surgical intensive care unit (SICU)** (if patient go to PACU): Location of the unit, expected length of stay, and visiting privileges

Preoperative Instructions:

- The patient is told when to stop fluid and food intake (**NPO**), usually after midnight the night before surgery.
- If surgery is scheduled for the afternoon, **clear liquids in the early morning**
- **Brush teeth**
- Special preparations, such as an **enema**, to empty the bowel to reduce fecal contamination preoperatively and straining or distention postoperatively.
- **Instructions for postoperative care:** how to report their **pain level** using a pain rating scale, before surgery so the patient is alert when being taught and has time to learn.

Postoperative exercises are taught to decrease complications, include :

- **Deep breathing** helps prevent the development of atelectasis
- **Leg exercises** improve circulation & help prevent complications related to stasis of blood, such as emboli formation.
- **Coughing** moves secretions to prevent pneumonia

Preoperative Data Collection\ Nursing Assessment



Health History
and
Psychosocial
Assessment



Physical
Assessment



Laboratory
and
Diagnostic
Test Data

Subjective Data:

- **Demographic information:** Name, age, marital status, occupation, roles?
- **History of condition for which surgery is scheduled:** Why are you having surgery?
- **Medical history:** Any allergies, acute or chronic conditions, current medications, pain, or prior hospitalizations?
- **Surgical history:** Any reactions or problems with anesthesia? Previous surgeries?
- **Tobacco use:** How much do you smoke? Pack-year history (number of packs per day _ number of years)?
- **Alcohol use:** How often do you drink alcohol? How much?
- **Coping techniques:** How do you usually cope with stressful situations? Support systems?
- **Family history:** Hereditary conditions, diabetes, cardiovascular, anesthesia problems?

Objective Data:

- As (**Body system review**, vital signs , oxygen saturation , Height and weight)
- **Emotional status:** calm, anxious, tearful
- **Neurological:** ability to follow instructions
- **Skin:** color, warmth, bruises, lesions, turgor, dryness, mucous membranes
- **Respiratory:** infection: cough; breath sounds; chronic obstructive pulmonary disease; respiratory rate, pattern, and effort; barrel chest
- **Cardiovascular:** angina, MI, heart failure, hypertension
- **Gastrointestinal:** bowel sounds, date of last bowel movement, abdominal distention.....
- **Musculoskeletal:** deformities, weakness, ROM

Immediate Preoperative Care

- Make sure that the client is **wearing his or her ID band**.
- Record the **client's VS**,
- **Determine and document the client's fall risk status**
- Check that the client is wearing **two special blood identification** when possibility of receiving blood.
- Help the client with **bathing and other hygiene measures**.
- Remove any prostheses, braces, bandages, hair pins, contact **lenses, hearing aids, and glasses**.
- Remove the client's **jewelry and valuables**.
- Pull the hair back and cover it with a **surgical cap**
- **Remove partial dentures** and place them in a denture cup with clear water.
- **Remove makeup**, and nail polish & artificial nails
- Make sure the **preoperative checklist in the client's health record is complete and signed**

Preoperative medications may be given to:

1

Reduce anxiety
and promote a
restful state

2

Decrease
secretion of
mucus and other
body fluids

3

Respond nausea
and reduce
emesis

4

Enhance the
effects of the
anesthetic



Informed Consent

- **Informed consent** is the patient's independent decision about whether to undergo a surgical procedure.
 - Before surgery, the client must sign a surgical consent form or operative permit.
 - If an adult client is confused, unconscious, a family member or protector must sign the consent form.
 - If the client is younger than 18 years of age, a **parent** or legal guardian must sign the consent form.
 - it is the surgeon's responsibility to provide a clear and simple explanation of what the surgery benefits, alternatives, possible risks, complications, disability, and removal of body parts as well as what to expect in the early and late postoperative periods.
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Preoperative Preparation

—1. Physical Preparation.

- **Skin preparation**, following **hospital policy**, to **prevent post-operative wound infection**. shaving hair and clean with antiseptic solution 10cm round surgical site then covered by sterile dressing
- **Elimination**, indwelling catheter and **enema** as physician orders
- **Food and fluids**, fasting based on type of anesthesia
- **Care of valuables**, Clothing/ grooming

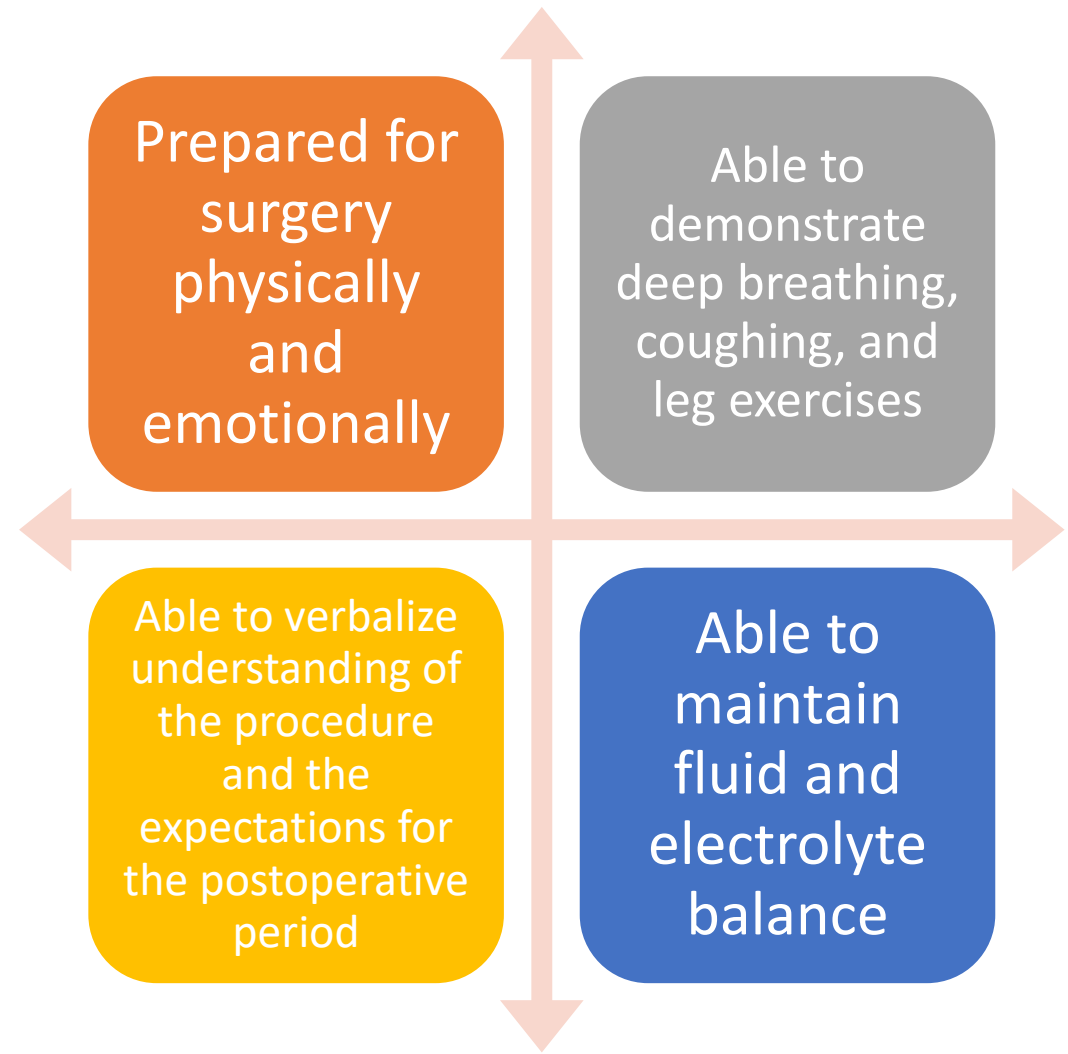
2. Psychosocial Preparation.

- —Careful preoperative teaching can **reduce fear and anxiety of the clients**.

Common preoperative problem statements include:

- **Anxiety** due to the surgical experience and outcome
 - **Fear** due to the potential for death, effects of impending surgery, or loss of control due to anesthesia
 - **Insufficient knowledge** of preoperative and postoperative routines
 - **Insomnia** due to stress or unfamiliar environment
 - **Limited coping** ability due to lack of problem-solving skills or adequate support

General nursing goals for all preoperative patients



- Identification band present: _____
2. Informed consent signed: _____ Special permits signed: _____
3. Surgical site: _____ (Ex: Sterilization)
4. History & physical examination report present: _____ Date: _____
5. Laboratory records present: _____
- CBC: _____ Hgb: _____ Urinalysis: _____ Hct: _____

- | 6. Item | Present | Removed |
|--|-------------|-------------|
| a. Natural teeth | _____ | _____ |
| Dentures; upper, lower, partial | _____ | _____ |
| Bridge, fixed; crown | _____ | _____ |
| b. Contact lenses | _____ | _____ |
| c. Other prostheses—type: _____ | _____ | _____ |
| d. Jewelry: | | |
| Wedding band (taped/tied) | _____ | _____ |
| Rings | _____ | _____ |
| Earrings: pierced, clip-on | _____ | _____ |
| Neck chains | _____ | _____ |
| Any other body piercings | _____ | _____ |
| e. Make-up | _____ | _____ |
| Nail polish | _____ | _____ |
| 7. Clothing | | |
| a. Clean patient gown | _____ | _____ |
| b. Cap | _____ | _____ |
| c. Sanitary pad, etc. | _____ | _____ |
| 8. Family instructed where to wait? _____ | | |
| 9. Valuables secured? _____ | | |
| 10. Blood available? _____ Ordered? _____ Where? _____ | | |
| 11. Preanesthetic medication given: _____ | Type: _____ | Time: _____ |
| 12. Voided: _____ Amount: _____ Time: _____ Catheter: _____ | | |
| Mouth care given: _____ | | |
| 13. Vital signs: Temperature: _____ Pulse: _____ Resp: _____ Blood Pressure: _____ | | |
| 14. Special problems/precautions: (Allergies, deafness, etc.): _____ | | |

Intraoperative

The Surgical Team:



The surgeon is the head of the surgical team, determining the preoperative diagnosis, performing the surgical procedure, the explanation of the risks and benefits, obtaining informed consent and postoperative management

Surgeon assistant assists surgeon in performing homeostasis, tissue retraction, and wound closure).

Anesthesiologist:
Anesthesia provider.

Scrub nurses: Role and responsibilities



Checks prior to beginning of the operating list (in collaboration with circulating nurses):



Cleanliness of the theatre environment



Temperature and humidity levels, and efficient air conditioning.



Preparation of instrument sets and accessories:



Scrubbing-up procedure: wash hands and arms, don sterile gown and gloves



Preparation of instrument trolley and Mayo stand



Documentation (electronic and hard copy)



Cleaning between surgical procedures and at the end of the operating list

- Swabs
- Needles
- Extra instruments
- Accessories (e.g. sutures and dressings)
- Passing instruments to the surgeon
- Anticipation of the surgeon 's needs
- Maintenance of the sterile field
- Swab, needle and instrument checks
- Handling and passing of instruments
- Specimens



Circulating nurses: Role and responsibilities

- ☐ Assisting the scrub practitioner in the preparation of the instrument sets, accessories, swabs, needles and instruments
- ☐ Preparation of sterile surgical gowns and gloves
- ☐ Tying donned sterile gowns
- ☐ Opening instrument sets
- ☐ Passing accessories to the scrub practitioner
- ☐ Maintenance of the sterile field
- ☐ Swab, needle and instrument checks with the scrub practitioner
- ☐ Anticipation of the scrub practitioner 's needs
- ☐ Documentation (electronic and hard copy)
- ☐ Cleaning between surgical procedures and at the end of the operating list
- Sterile techniques are always maintained.

Intra operative team must be

Applying surgical hand scrub

Gloves are worn

Enhance aseptic technique.

Wears shoe covers, caps, masks, and goggles

Surgical case cart preparation

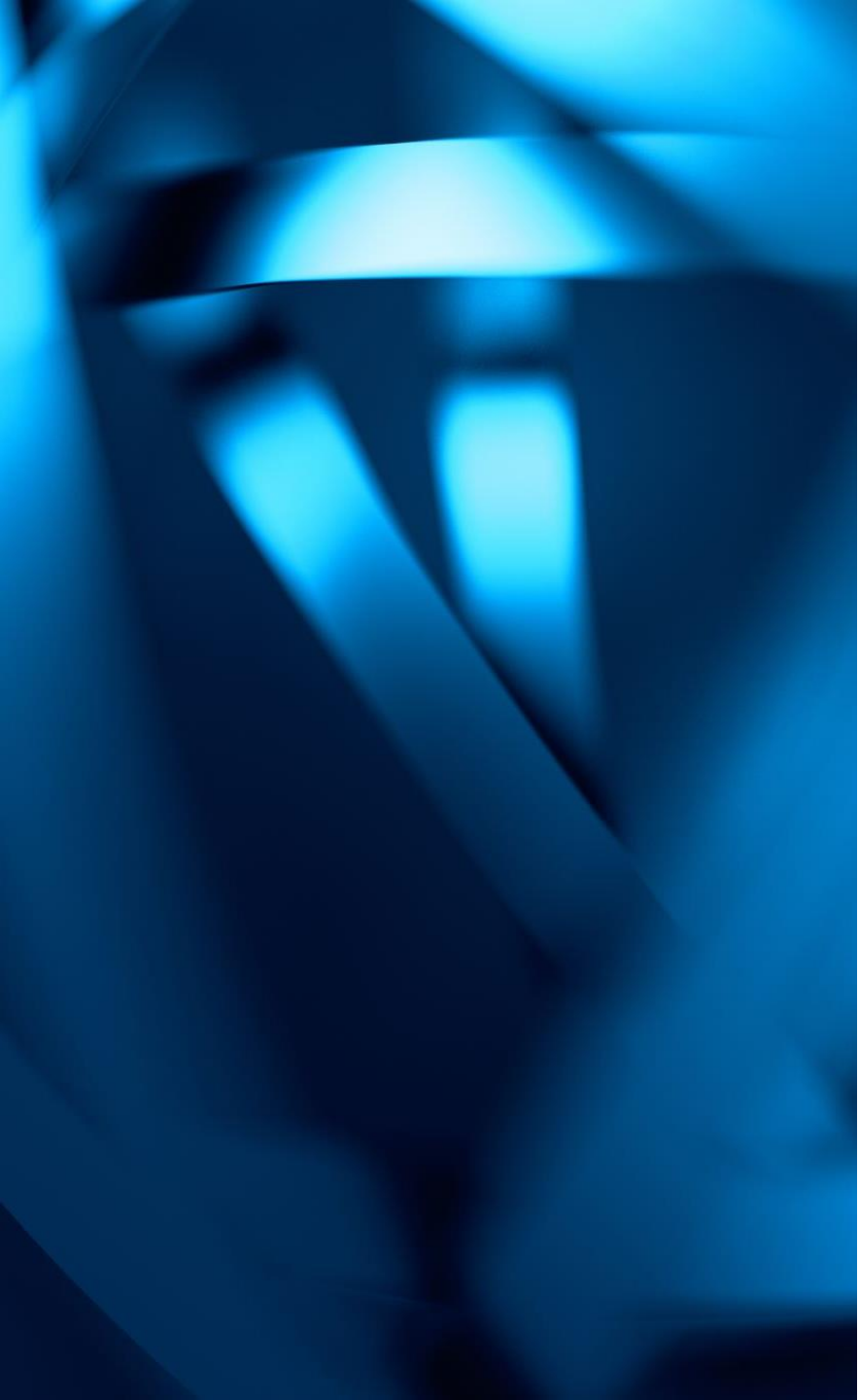
Count needles and sponges & others surgical instruments

Ensure safety, electrical equipment is checked

Anesthesia

- Agent used to alter sensation so that surgery can be done painlessly & safely

Type of anesthesia	Description
General	• Blocks awareness centers in the brain • Produces unconsciousness, body relaxation, and loss of sensation • Is administered by inhalation or I.V. infusion
Regional	• Inhibits excitatory processes in nerve endings or fibers • Provides analgesia over a specific body area • Doesn't produces unconsciousness • It is administered by nerve block, I.V. regional block with tourniquet, spinal (intrathecal) block, or epidural block
Local	• Blocks nerve impulse transmission at the site of action • Provides analgesia over a limited area • Doesn't produces unconsciousness • Is administered topically or by infiltration



There are three stages of general anesthesia:

Induction: Unconsciousness is induced.

Maintenance: Period during which the surgical procedure is performed.

Emergence: Surgery is completed, and the patient is prepared to return to consciousness; neuromuscular blocking agents are reversed.



The Operating Room Environment

- ☞ The OR environment is physically isolated from other areas of the hospital, air is filtered, and positive pressure is maintained to reduce the number of possible microbes that can cause infection.
- ☞ OR are designed to be efficient, in that the needed equipment and supplies are immediately available for use
- ☞ Usually, the furniture is made of stainless steel for easy cleaning and disinfection.
- ☞ The temperature in the OR is a cooler environment that does not promote bacterial growth, more comfort for OR personnel working in bright lights and wearing and enhances client comfort and safety.

Prevention of Intraoperative Complications

1.Infection:

- ☞ Strict aseptic techniques are necessary before and during surgery.
- ☞ If a nurse notes a break in technique, immediately notifies the surgeon

2.Fluid volume excess or deficit:

The circulating nurse is responsible for recording and keeping a running total of IV fluids administered

3.Injury related to positioning:

- The OR staff positions the client on the OR table according to the type of surgery.

4.Hypothermia:

- The client may be at risk for hypothermia related to the low temperature in the
- The OR administration of cold IV fluids, inhalation of cool gases, and exposure of body surfaces for the surgical procedure, opened incisions/wounds, and prolonged inactivity.
- For some surgeries, the body temperature is purposely lowered to make the procedure safer

5.Malignant hyperthermia (MH)

- Occurs when body temperature, muscle metabolism, and heat production increase rapidly, progressively, and uncontrollably in response to stress and some anesthetic agents.



The circulating nurse

Closely monitors the client for signs of hyperthermia.

If the client 's temperature begins to rise rapidly, anesthesia is discontinued and the OR team implements measures to correct physiological problems, such as dysrhythmias.

Potential Adverse Effects of Anesthesia

Cardiac dysrhythmia from electrolyte imbalance or adverse effect of anesthetic agents

Myocardial depression, bradycardia, and circulatory collapse

Central nervous system agitation, **seizures**, and **respiratory** arrest

Over sedation or under sedation

Agitation or disorientation, especially in elderly patients

Hypoxemia or hypercarbia from hypoventilation and inadequate respiratory support during anesthesia

Laryngeal trauma, oral trauma, and broken teeth from difficult intubation

Postoperative

Transport of the Client

- ▮ Immediately after the surgical procedure is complete, the client is transported to the post-anesthesia care unit (PACU), known as the post-anesthesia recovery room.
- ▮ The nursing staff there is specifically **knowledgeable** in the care of clients recovering from anesthesia.
- ▮ Specialized **equipment** is available to **monitor** and treat the client.
- ▮ Patients may remain in a PACU unit for as long as **4 to 6 hours**, depending on the type of surgery and any preexisting conditions
- ▮ The nurse receiving the client from the OR needs the following information

- Medical diagnosis and surgical procedure done
- Past medical history and allergies
- Age, general condition, airway status, and current vital signs
- Anesthetic agents and medications given during surgery
- Complications during surgery
- Any pathology found and if so whether family members are informed
- Amounts of fluids & blood administered or lost
- Any tubes connected

Immediate Postoperative Initial Assessment:

Initial Assessment

- ▮ Airway patency
- ▮ Effectiveness of respiration
- ▮ Presence of artificial airways
- ▮ Mechanical ventilation, or supplemental oxygen
- ▮ Circulatory status, vital signs
- ▮ Wound condition, including dressings and drains

- ▮ Fluid balance, including IV fluids, output from catheters and drains and ability to void
- ▮ Level of consciousness and pain
- ▮ The nurse 's major responsibilities during the client 's PACU are to
 - ✓ ensure a patent airway
 - ✓ help maintain adequate circulation
 - ✓ prevent or assist with the treatment of shock
 - ✓ maintain proper position and function of drains, tubes, and IV infusions; monitor for potential complications.

Modified Aldrete Scale for Assessing Recovery from Anesthesia

Score	0	1	2
Activity	Unable to move extremities voluntarily or on command	Able to move two extremities Voluntarily or on command	Able to move all extremities voluntarily or on command
Respiration	Apneic	Dyspnea or limited breathing	Able to breathe deeply and cough freely
Circulation	BP+/-50mmHg of Pre-anesthesia level	BP+/- 20-49mmHg of Pre-anesthesia level	BP+/- 20mmHg of Pre-anesthesia level
Consciousness	Unresponsive	Arousable with verbal stimuli	Fully awake
SpO2	< 90 % with supplemental oxygen	Needs supplemental oxygen to maintain > 90%	> 92% on room air

Immediate nursing interventions

- Assess breathing and administer supplemental oxygen, if prescribed.
- Monitor vital signs and note skin warmth, moisture, and color.
- Assess the surgical site and wound drainage systems.
- Assess level of consciousness, orientation, and ability to move extremities.
- Connect all drainage tubes to gravity or suction as indicated and monitor closed drainage systems.
- Assess pain level, pain characteristics (location, quality) and timing, type, and route of administration of last pain medication.

- Administer analgesics as prescribed and assess their effectiveness in relieving pain.
- Position patient to enhance comfort, safety, and lung expansion.
- Assess IV sites for patency and infusions for correct rate and solution.
- Assess urine output in closed drainage system or the patient's urge to void and bladder distention.
- Place call light, emesis basin, ice chips (if allowed), and bedpan or urinal within reach.
- Provide information to patient and family

Measures are used to determine the patient's readiness for discharge from the PACU

Stable vital signs

Orientation to
person, place,
events, and time

Uncompromised
pulmonary
function

Pulse oximetry
readings indicating
adequate blood
Oxygen saturation

Urine output at
least 30 mL/h

Nausea and
vomiting absent or
under control

Minimal pain









Essay Question 1:

Discuss the role of the perioperative nurse in ensuring patient safety across the preoperative, intraoperative, and postoperative phases.

Sample Answer:

The perioperative nurse plays a vital role in safeguarding patient safety throughout the surgical experience. In the preoperative phase, the nurse conducts thorough assessments, verifies informed consent, reviews medical history, and educates the patient about the procedure and recovery expectations. During the intraoperative phase, the nurse maintains a sterile environment, assists the surgical team, monitors the patient's vital signs, and participates in the surgical safety checklist and time-out procedures to prevent wrong-site surgery. Postoperatively, the nurse ensures airway patency, monitors for complications such as bleeding or infection, manages pain, and facilitates early mobilization to prevent deep vein thrombosis. Effective communication, vigilance, and adherence to protocols are essential throughout all phases to promote optimal outcomes.

Essay Question 2:

Explain the importance of preoperative patient education and how it impacts postoperative recovery.

Sample Answer:

Preoperative education is a cornerstone of perioperative nursing care. It prepares patients mentally and physically for surgery, reduces anxiety, and enhances cooperation. Educating patients about fasting guidelines, medication adjustments, pain management strategies, and postoperative expectations empowers them to participate actively in their recovery. Studies show that informed patients experience fewer complications, shorter hospital stays, and better adherence to postoperative instructions. For example, teaching the use of incentive spirometry and early ambulation helps prevent pulmonary complications and promotes faster healing. Ultimately, preoperative education fosters trust, improves satisfaction, and contributes to safer surgical outcomes.

- <https://youtu.be/HheuG5REcKw>
- <https://www.youtube.com/watch?v=rNtDIY0TS6U>

thank
you

